Defuse Game

Game Overview
Two player co-operative puzzle game revolving around a physical controller. The main idea is to use a physical controller which player 1 operates which will interact with the gameworld being navigated by player 2. The players will have a pre-defined amount of time to solve puzzles, navigate the gameworld and work together to “defuse the bomb”.

Art Vision
The visual design of the game is not wholly decided yet, but a cartoon/non-realistic art style is likely. During gameplay, the player 2 gameworld will be calm however as time reduces, the colour palette will shift and music might change to indicate the need for quick action.

Core Gameplay
The core gameplay revolves around solving numerous sub-objectives (puzzles) to unlock/achieve the final goal of defusing the bomb. These objectives will have differing layouts and ideas, for instance a platform section for player 2 and a digital screen memory game for player 1.

Technology and Development Approach
The game will be developed using Unity and programmed in C#. A mixed asset approach will be used. All core components like audio, gameworld, puzzles, interaction with hardware would be implemented from scratch. There will be use of some third-party assets for user interface designs and visual effects. As for the hardware, this will be created from scratch most likely 3D printed.

Flagship Technology and Innovation
The main technology is a hybrid software/hardware setup, with hardware communicating in real-time to affect the virtual world.

Unlike many co-operative games where both players interact within the same environment, in this game the gameplay loop is split in half, the gameplay experience being very different for each player, allowing people to choose the gamestyle they prefer. The innovative part is how hardware is integrated into gameplay. The gameworld will shift for player 2 depending on the actions of player 1, causing the necessity for effective and clear communication, something that may be hard to do in a timed environment.

Computer Science Contribution
Computer Science which will be used is real-time hardware/software communication. The game logic is defined and executed through multiple systems working together concurrently alongside in-game events.

Testing and Evaluation
Testing will be done through iterative playtesting with peers and students. Various puzzles and timings will be tested to find the best mix. The feedback will focus on the clarity of puzzles and

how enjoyable it is for the hardware player. Quantitative data will be combined with qualitative feedback to guide improvements.

MVP Goals

- Simple controller that interacts with the game world with up to 3 components
- Simple puzzle/platforming in the game that requires cooperation between the players to complete

Beta Goals

- Complete level with multiple digital puzzles and interactivity between the two players
- Multiple independent and cooperative hardware puzzles
- Initial design for the controller

Final Goals

- Finished controller with several minigames and complete design
- Multiple levels/rooms in the game world with coop puzzles

Music Game

Game Overview
This is a Rhythm based bullet-hell game where the player fights with a single boss that is created using music. The idea here is making music an important part of the game, not just playing it as background music. Boss attacks, visuals and difficulty will change depending on the music. Instead of having more levels and enemies, the game is a boss fight that evolves as the song changes.

Art Vision
The visual design is based on the concept of the communication between the gameplay and music. Rhythm and intensity are closely connected with visuals and motion which can assist players to comprehend when to act and when there is danger and may score points. The music, graphics of the arena, boss and fight change concurrently to create a unified experience to the gamer throughout the game.

Core Gameplay
The game is one boss fight and continues till the end of the song. The player then picks the difficulty level, depending on the type of music he has picked in the playlist which pre-determines the actions of the boss. The battle is divided into rounds according to various aspects of the music. To begin with, the music is simple and the boss attacks are slow and easy to read. However in time, additional layers of music are added and the attacks become faster and are going to be more complicated. There are no health bars or scripted boss phases. The player must dodge shots and attack the boss in line with the music that increases the player's score. The main goal of the game is to score more by getting combos, avoiding mistakes and making actions in rhythm with the music. The fight ends when the song is finished and an Emperor will then assess the performance by the use of the scores.

Technology and Development Approach
It is going to be created in Unity and C#. The plan would be based on the mixed asset strategy. The strategy would follow the mixed asset approach. All core components like audio analysis, bullet pattern, boss moves, scores, would be implemented completely from scratch. There could be the use of some third-party assets for user interface designs and effects.

Flagship Technology and Innovation
The main technology is real-time, music-driven combat systems. The system analyses music data and uses it to control boss attacks, visuals and difficulty. Unlike many rhythm games where patterns are fixed, in this game the pattern is determined by the song itself. Every song creates different fights which increases replay value. The innovative part is how closely music is linked to gameplay. The attack speed, difficulty, and graphics change depending on the music being played when a player is in action.

Computer Science Contribution
Computer Science will be used for real-time audio processing, procedurally generated bullet patterns, adaptive difficulty level and scoring algorithms. The game logic is defined and

executed through multiple systems working together concurrently at runtime and not events triggered in games.

Testing and Evaluation
Testing will be done through iterative playtesting with peers and students. Various categories of music will be used to check efficiency across all genres. The feedback will focus on the clarity of bullet patterns, how well the rhythm communication is and overall experience. Quantitative data will be combined with qualitative feedback to guide improvements

MVP Goals

- Initial design of an arena
- Simple attacks based on the bpm/volume of a given song
- An attempt at getting a difficulty score from any song

Beta Goals

- More in depth design of an arena
- Separating elements of a song for manipulation
- Multiple attacks based on elements in the track
- A simple scoring system for players' performance
- More developed difficulty scoring system

Final Goals

- A finished arena with themes based on the genre of music
- Given a playlist, gives the difficulty rating of each song in the playlist
- Fully developed scoring system based on the player performance and the difficulty of the song
- Multiple attacks and stages of the fight with song elements being introduced at each stage